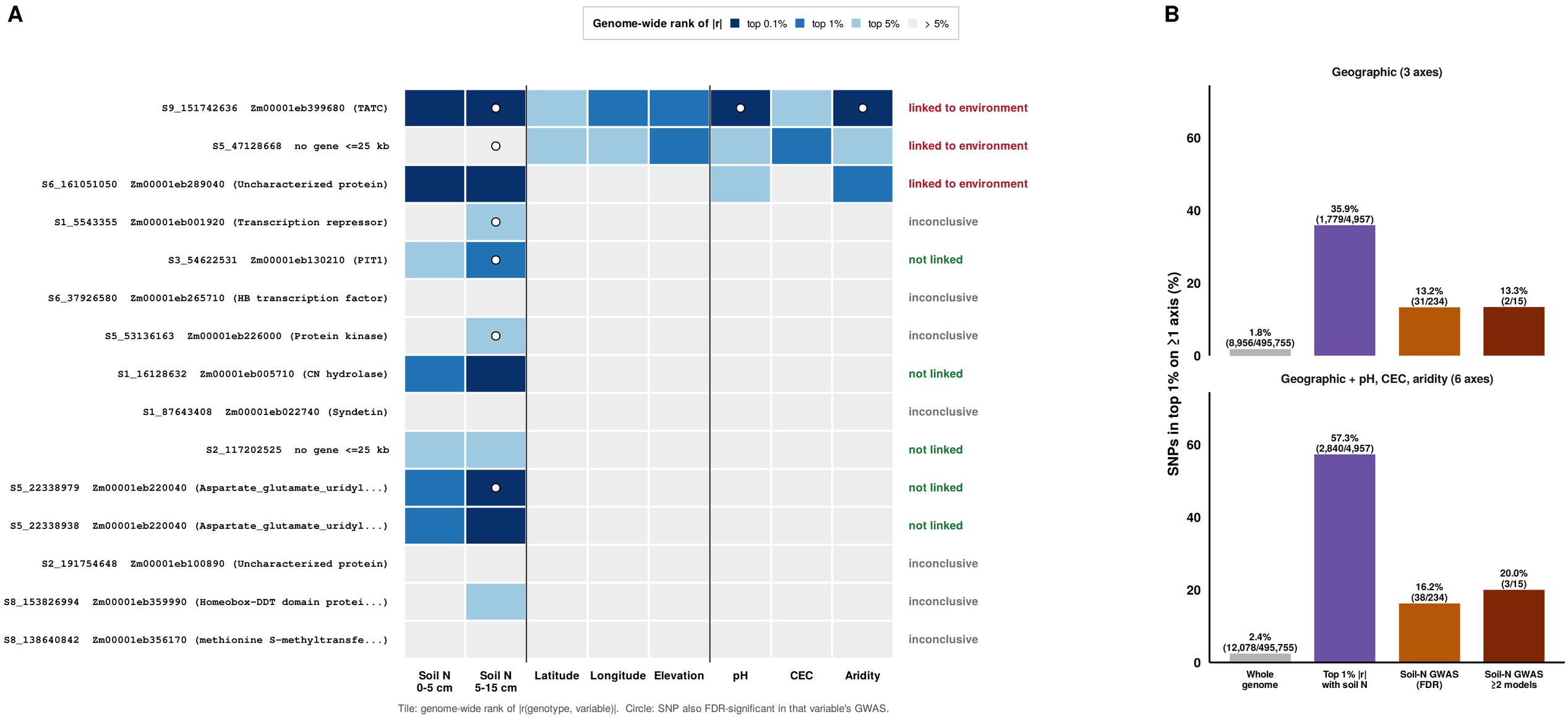
A
Genome-wide rank of |r|
top 0.1%
top 1%
top 5%
> 5%
S9_151742636 Zm00001eb399680 (TATC)
S5_47128668 no gene <=25 kb
S6_161051050 Zm00001eb289040 (Uncharacterized protein)
S1_5543355 Zm00001eb001920 (Transcription repressor)
S3_54622531 Zm00001eb130210 (PIT1)
S6_37926580 Zm00001eb265710 (HB transcription factor)
S5_53136163 Zm00001eb226000 (Protein kinase)
S1_16128632 Zm00001eb005710 (CN hydrolase)
S1_87643408 Zm00001eb022740 (Syndetin)
S2_117202525 no gene <=25 kb
S5_22338979 Zm00001eb220040 (Aspartate_glutamate_uridyl...)
S5_22338938 Zm00001eb220040 (Aspartate_glutamate_uridyl...)
S2_191754648 Zm00001eb100890 (Uncharacterized protein)
S8_153826994 Zm00001eb359990 (Homeobox-DDT domain protei...)
S8_138640842 Zm00001eb356170 (methionine S-methyltransfe...)
linked to environment
linked to environment
linked to environment
inconclusive
not linked
inconclusive
inconclusive
not linked
inconclusive
not linked
not linked
inconclusive
inconclusive
inconclusive
Soil N 0-5 cm
Soil N 5-15 cm
Latitude
Longitude
Elevation
pH
CEC
Aridity
Tile: genome-wide rank of |r(genotype, variable)|. Circle: SNP also FDR-significant in that variable's GWAS.
B
Geographic (3 axes)
Geographic + pH, CEC, aridity (6 axes)
SNPs in top 1% on ≥1 axis (%)
0
20
40
60
1.8% (8,956/495,755)
35.9% (1,779/4,957)
13.2% (31/234)
13.3% (2/15)
2.4% (12,078/495,755)
57.3% (2,840/4,957)
16.2% (38/234)
20.0% (3/15)
Whole genome
Top 1% |r| with soil N
Soil-N GWAS (FDR)
Soil-N GWAS ≥2 models